

# Ernesto Miró Peraza

## Senior Software Engineer | Full-Stack .NET/Angular

Cloud architecture · Technical leadership · Artificial intelligence-assisted engineering

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### PROFESSIONAL SUMMARY

Computer Science Engineer with 9 years of experience delivering enterprise software solutions. Specialized in C#/.NET and Angular, with experience in multi-tenant SaaS (Software as a Service) platforms, microservices, event-driven architecture, identity, security, and integration across GCP (Google Cloud Platform) and Azure. Combines hands-on development, architecture definition, technical leadership, and delivery coordination. Also works on AI application capabilities using Semantic Kernel, language models, retrieval-augmented generation (RAG), and agentic workflows to connect business information and application tools.

### AREAS OF EXPERTISE

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| <b>Backend &amp; Architecture</b><br>C#/.NET · ASP.NET Core · web APIs · GraphQL · microservices · event-driven systems · multi-tenant SaaS | <b>Cloud &amp; Security</b><br>GCP · Azure · OAuth 2.0 · OpenID Connect · identity federation · access control · Terraform · Docker                        |
| <b>Frontend &amp; Data</b><br>Angular · TypeScript · Angular Material · RxJS · Signals · SQL Server · PostgreSQL · Firestore · BigQuery     | <b>Engineering &amp; Leadership</b><br>SOLID principles · Clean Code · Scrum · Azure DevOps · Jira · pull request review · agentic artificial intelligence |
| <b>AI Application Development</b><br>Semantic Kernel · LLM integration · RAG · agents and tools · local inference                           | <b>AI-Assisted Engineering</b><br>Codex and Claude Code in Visual Studio Code · project context · planning, implementation, and human review               |

### PROFESSIONAL EXPERIENCE

#### Full-Stack Software Engineer & Technical Lead

##### Private Automotive-Sector Client | Mexico | Remote | 2026

Independent professional project to design and develop an enterprise CRM (Customer Relationship Management) platform.

- Lead a 3-person team, coordinating frontend development and QA (Quality Assurance), while owning architecture, infrastructure, and client communication.
- Designed the decoupled architecture and lead development using Angular 21, TypeScript, .NET 9, ASP.NET Core Web API, Entity Framework Core 9, and Microsoft SQL Server.
- Implemented modules for customers, vehicles, services, quotations, automotive glass, appointments, users, companies, and branches, with multi-company and multi-branch context.
- Built an appointment calendar with working schedules, holidays, per-slot capacity, availability rules, and conflict validation.
- Implemented authentication using JWT (JSON Web Token) in HttpOnly cookies, authorization, session control, validation, and centralized exception handling.
- Manage planning, task assignment, progress reviews, and technical blocker resolution.

## PROFESSIONAL EXPERIENCE

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### Senior Full-Stack Software Engineer

**Itopia | United States | Remote | 2019-2026**

U.S.-based SaaS company serving K-12 education, focused on cloud platforms, virtualization, and remote access.

#### Technical Leadership & Architecture

- Provided internal technical leadership for a small team consisting of one frontend engineer, one backend engineer, and one QA specialist.
- Created and assigned tasks, estimated work, reviewed PRs (Pull Requests), guided implementation, and coordinated delivery across frontend, backend, and QA.
- Orchestrated the architecture of IGW (Itopia Gateway), defining domains, integrations, conventions, and technical criteria for its evolution.
- Standardized structured logging, dependency injection, secret management, and code conventions across more than 5 repositories.
- Designed AI (Artificial Intelligence)-assisted engineering workflows using skills, subagents, MCP (Model Context Protocol), and documentation as context.

#### Next-Generation Multi-Tenant SaaS Platform | 2025-2026

- Designed and implemented microservices with .NET 10, gRPC (Google Remote Procedure Call), and REST (Representational State Transfer) APIs for a multi-tenant SaaS platform serving school districts.
- Built a single-entry gateway with OAuth 2.0 client credentials, scoped authorization, and rate limiting, along with session, organization, catalog, and internal event-delivery domains.
- Developed automated tenant onboarding on Azure using state machines, Entra ID consent flows, callbacks, event fan-out, and scheduled retries.
- Designed capacity and cost management for Azure Virtual Desktop host pools using ARM (Azure Resource Manager), batch startup, and configurable buffers.
- Implemented long-running asynchronous operations, polling, ghost-session cleanup, feature flags, and idempotency.
- Identified authentication vulnerabilities and designed mitigations using strict audience validation and HMAC (Hash-Based Message Authentication Code) indexes for sensitive data.
- Defined infrastructure as code with Terraform and CI/CD (Continuous Integration and Continuous Delivery) pipelines using Cloud Build.

#### EdTech VDI Platform | 2019-2025

- Built and evolved a polyglot VDI (Virtual Desktop Infrastructure) platform of approximately 20 C#/.NET and Node.js/TypeScript services on Cloud Run and Cloud Functions.
- Developed rostering and identity integrations with Clever, ClassLink/OneRoster, Google Workspace, Google Admin Directory, Google Classroom API, Microsoft Graph, Entra ID, OAuth 2.0, and SAML 2.0.
- Implemented metering in BigQuery, virtual-machine orchestration in Compute Engine, and an event engine with Pub/Sub, delay and retry queues, callbacks, and an SSE (Server-Sent Events) gateway.
- Built Angular portals for administration, end users, and support, with typed GraphQL, PWA (Progressive Web Application) capabilities, and remote-desktop viewer integration.
- Developed a .NET Windows agent for VDI virtual machines, including inactivity detection, event publishing, and instance lifecycle management.
- Implemented envelope encryption with Cloud KMS, secret rotation, Terraform, and Cloud Build.
- Developed platform-wide expertise in the WebRTC/GStreamer streaming stack to analyze, diagnose, and evolve it, without claiming authorship of its foundational components.
- Contributed to WorkAnyWhere, Cloud Compliance Manager, CloudApps, and ItopiaOne.

EARLIER EXPERIENCE

Project Specialist  
XETID | Cuba | 2017-2019

- Developed enterprise systems in the Entity Management Division and contributed to the Distra ARP project.
- Implemented and maintained backend and frontend features using PostgreSQL, Zend Framework, Ext JS, and Bootstrap.
- Analyzed requirements and resolved functional and technical needs within a multidisciplinary team.

Software Engineer - Academic Project  
University of Information Sciences (UCI) | Cuba | 2016-2017

- Developed SBRSPPLIT, a system that predicts compatibility between students and courses based on individual aptitudes.
- Implemented the solution using Python, PyQt5, and PostgreSQL, including Excel file processing and packaging with PyInstaller.

SELECTED PERSONAL PROJECT

Personal AI Project — Conversational Specialist Search

Designed the conversational workflow and developed retrieval and tool-calling components using Semantic Kernel and RAG over a vectorized specialist catalog.

The workflow design connects a problem description with relevant specialist profiles, uses an LLM to explain matches, and incorporates user clarification to refine retrieval.

TECHNOLOGIES & PRACTICES

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|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Backend                         | C#/.NET (.NET Framework 4.7.2 through .NET 10), ASP.NET Core, Node.js, TypeScript, Python, REST (Representational State Transfer), gRPC (Google Remote Procedure Call), GraphQL, Apollo Server, and SSE (Server-Sent Events).                                                                                                                                           |
| Architecture                    | Microservices, event-driven architecture, multi-tenant SaaS, API Gateway, state machines, sagas, callbacks, fan-out, webhooks, retries, and idempotency.                                                                                                                                                                                                                |
| Cloud                           | GCP (Google Cloud Platform): Cloud Run, Cloud Functions, Pub/Sub, Cloud Tasks, Cloud Scheduler, Firestore, BigQuery, Cloud KMS, Secret Manager, Compute Engine, and Cloud Build. Azure: Azure Virtual Desktop, ARM (Azure Resource Manager), Entra ID, Microsoft Graph, MSAL (Microsoft Authentication Library), and Managed Identity. Terraform and Docker.            |
| Security                        | OAuth 2.0, OIDC (OpenID Connect), SAML 2.0 (Security Assertion Markup Language), JWT (JSON Web Token), JWE (JSON Web Encryption), JWKS (JSON Web Key Set), Firebase Authentication, SSO (Single Sign-On), RBAC (Role-Based Access Control), scoped authorization, envelope encryption, HMAC (Hash-Based Message Authentication Code) blind indexing, and rate limiting. |
| Frontend & Data                 | Angular, Angular Material, RxJS, Signals, reactive forms, standalone components, GraphQL with code generation, PWA, SQL Server, PostgreSQL, Firestore, BigQuery, Redis, and Entity Framework Core.                                                                                                                                                                      |
| Engineering                     | SOLID principles, object-oriented programming, Clean Code, design patterns, refactoring, code review, structured logging, feature flags, Git, GitHub, Scrum, Azure DevOps, and Jira.                                                                                                                                                                                    |
| Applied Artificial Intelligence | Claude Code, Codex, MCP (Model Context Protocol), skills, subagents, documentation as context, and backlog-to-pull-request workflow automation. Uses both tools through Visual Studio Code extensions and its integrated terminal to analyze, plan, implement, and review changes with project context and human oversight.                                             |
| AI Orchestration                | Semantic Kernel SDK, LLM Integration, AI Agents, Tool Calling / Function Calling, Agentic Workflows, and Multi-Agent Orchestration.                                                                                                                                                                                                                                     |
| Knowledge Retrieval             | Embeddings, Semantic Search, Vector Search, RAG Architecture, PostgreSQL + pgvector, Cosine Similarity, and Top-K Retrieval.                                                                                                                                                                                                                                            |
| Model Integration & Inference   | ONNX Runtime, Local Model Inference, Hugging Face Tokenization, Embedding Models, Transformer Models, and Model Artifacts.                                                                                                                                                                                                                                              |
| AI Application Patterns         | Human-in-the-Loop, Structured Outputs, Event-Driven Agents, Background Agents, Database Tools, and External API Tools.                                                                                                                                                                                                                                                  |
| AI Provider & SDK Integration   | OpenAI SDK, Azure OpenAI SDK, Anthropic SDK, Google Gemini SDK, and REST API Integration.                                                                                                                                                                                                                                                                               |

EDUCATION & LANGUAGES

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| <p>Computer Science Engineering</p> <p>University of Information Sciences (UCI), Cuba   2012-2017</p> | <p>Languages</p> <ul style="list-style-type: none"><li>• Spanish: Native.</li><li>• English: Professional working proficiency in conversations and meetings; advanced reading, writing, and technical vocabulary.</li></ul> |
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